Parity-Renormalized First-Alias Packets: Exact Signs, Clock Phase, and the Neighboring-Shell Interface

Bin Wang

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Abstract

The first counterloop alias and the square-root parity leakage occur at the same weighted order, so their signs and normalization cannot be suppressed. We restore the Hardy scaling hidden in the earlier first-alias ledger and split the coefficient defect exactly into a raw-trace packet, a signed parity packet, a finite-radius radial correction, and the roots-of-unity alias impulse. At the even first alias the parity correction is positive, whereas the counterloop defect is subtracted from the total residual. The parity packet has a uniform moving-order expansion with an explicit quadratic remainder. On a fixed clock phase η, its leading ratio to the alias defect is $C_* C_M \lambda^\eta$. Scalar balance is possible only at $\eta_* = 3.0609149137\ldots$; throughout the usual floor/ceil/nearest window $|\eta| \le 1$, parity supplies at most $0.3438880199\ldots$ of the alias. Thus scalar parity subtraction alone cannot close the first-alias equation. This is a scoped negative result: the raw physical boundary packet and the neighboring shell may still provide the missing signed contribution. We finish with a phase- and coordinate-preserving interface for that calculation. No joint full-trace replacement is proved.

1 Normalization and four packet types

Put

$$\lambda = 1.678573510428322\ldots, \qquad r_H = 0.85, \qquad R = 1.4, \qquad \beta = (r_H\sqrt{\lambda})^{-1}. \tag{1}$$

The number r_H is the chosen Hardy scaling radius, not a dynamical invariant. Let P_n be the deterministic physical flat trace and let K_σ be the row-normalized noisy Markov operator. The peripheral parity eigenvalue is written

$$\lambda_-(\sigma) = -(1-\delta_\sigma), \qquad \delta_\sigma = C_*\sqrt{\sigma} + o(\sqrt{\sigma}), \qquad C_* = 0.105258535936908\ldots. \tag{2}$$

The decimal evaluates an archived positive integral; positivity and the square-root law, rather than the rounded decimal, are the analytic inputs.

We use explicit Hardy superscripts to avoid mixing two archived conventions:

$$c^H_{\sigma,n} = r_H^{-n}\{\operatorname{Tr} K_\sigma^n - 1 - \lambda_-(\sigma)^n\}, \tag{3}$$

$$c^H_n = r_H^{-n}\{P_n - 1 - (-1)^n\}. \tag{4}$$

Define the raw-trace and parity packets

$$\mathcal{T}_{\sigma,n} = r_H^{-n}\{\operatorname{Tr} K_\sigma^n - P_n\}, \tag{5}$$

$$\mathcal{P}_{\sigma,n} = r_H^{-n}\{(-1)^n - \lambda_-(\sigma)^n\} = (-1)^n r_H^{-n}\{1-(1-\delta_\sigma)^n\}. \tag{6}$$

Then, identically,

$$c^H_{\sigma,n} - c^H_n = \mathcal{T}_{\sigma,n} + \mathcal{P}_{\sigma,n}. \tag{7}$$

For the finite counterloop let

$$s_{k,n} = \beta_k^n\big(2k\mathbf{1}_{2k|n} - 1 - (-1)^n\big), \qquad \beta_k \longrightarrow \beta. \tag{8}$$

Rename the limiting pole moment and numerator anchor to avoid collisions with older uses of p_n and a_n:

$$p_n^{\text{pole}} = -2\mathbf{1}_{2|n}\beta^n, \qquad a_n^{\text{num}} = c_n^H - p_n^{\text{pole}}. \tag{9}$$

2 Exact alias and parity algebra

Theorem 2.1 (Radial-plus-alias decomposition)**.** *For every $k, n \geq 1$, the counterloop defect is exactly*

$$\boxed{\mathcal{A}_{k,n} := s_{k,n} - p_n^{\text{pole}} = 2\mathbf{1}_{2|n}(\beta^n - \beta_k^n) + 2k\beta_k^n\mathbf{1}_{2k|n}.} \tag{10}$$

The first term is finite-radius radial drift and the second is the exact roots-of-unity alias impulse. In particular, $\mathcal{A}_{k,n} = 0$ for every odd n, while

$$\mathcal{A}_{k,2k} = (2k-2)\beta_k^{2k} + 2\beta^{2k} > 0. \tag{11}$$

Proof. Substitute (8) and (9). If n is odd, both $1 + (-1)^n$ and the two divisibility indicators vanish. If n is even, the non-alias part is $-2\beta_k^n + 2\beta^n$, and an order divisible by $2k$ receives the additional impulse $2k\beta_k^n$. Setting $n = 2k$ gives (11). □

The sign ledger is now unambiguous:

order	parity packet $\mathcal{P}_{\sigma,n}$	counterloop defect $\mathcal{A}_{k,n}$	signs in the residual
odd	negative	0	$+\mathcal{P}$
even, $2k \nmid n$	positive	$2(\beta^n - \beta_k^n)$	$+\mathcal{P} - \mathcal{A}$
first alias $n = 2k$	positive	$(2k-2)\beta_k^{2k} + 2\beta^{2k}$	$+\mathcal{P} - \mathcal{A}$

Only the parity sign is asserted in the middle row; the radial difference need not be assigned a sign without a finite-k radius inequality.

Theorem 2.2 (Parity-renormalized first-alias packet identity)**.** *Let*

$$e_{\sigma,k,n} = c_{\sigma,n}^H - s_{k,n} - a_n^{\text{num}}. \tag{12}$$

Then at the first alias,

$$\begin{aligned} e_{\sigma,k,2k} &= \mathcal{T}_{\sigma,2k} + \mathcal{P}_{\sigma,2k} - \mathcal{A}_{k,2k} \qquad (13)\\ &= r_H^{-2k}\{\operatorname{Tr} K_\sigma^{2k} - P_{2k} + 1 - \lambda_-(\sigma)^{2k}\} \\ &\qquad - \{(2k-2)\beta_k^{2k} + 2\beta^{2k}\}. \qquad (14) \end{aligned}$$

In particular, the scalar parity contribution in (14) *is*

$$r_H^{-2k}\{1 - \lambda_-(\sigma)^{2k}\} = r_H^{-2k}\{1 - (1-\delta_\sigma)^{2k}\} > 0. \tag{15}$$

Proof. Using $a_n^{\text{num}} = c_n^H - p_n^{\text{pole}}$ in (12) gives

$$e_{\sigma,k,n} = (c_{\sigma,n}^H - c_n^H) - (s_{k,n} - p_n^{\text{pole}}).$$

Apply (7), theorem 2.1, and set $n = 2k$. Equation (15) follows because $0 < 1 - \delta_\sigma < 1$ for sufficiently small noise. □

This theorem is an exact coefficient identity. It does not approximate $\mathcal{T}_{\sigma,2k}$ by a local Gaussian law and therefore does not assume the missing trace-observation theorem.

3 Uniform parity packet on the first-alias clock

Theorem 3.1 (Uniform scalar parity expansion)**.** *For $0 \le \delta < 1$ and every $n \ge 1$,*

$$\left|\mathcal{P}_{\sigma,n} - (-1)^n n\delta r_H^{-n}\right| \le \frac{n(n-1)}{2}\delta^2 r_H^{-n}, \tag{16}$$

where the left side uses (6) *with $\delta_\sigma = \delta$. Consequently, if*

$$k_\sigma = \frac{\log(1/\sigma)}{2\log\lambda} + O(1), \tag{17}$$

then

$$\mathcal{P}_{\sigma,2k_\sigma} = 2k_\sigma C_* \sqrt{\sigma}\, r_H^{-2k_\sigma}\{1+o(1)\}. \tag{18}$$

Its weighted size and the weighted first-alias defect share the growth exponent

$$\chi = \frac{\log(\beta R)}{\log\lambda} = 0.463406944517002\ldots. \tag{19}$$

Proof. Taylor's theorem for $f(\delta) = 1-(1-\delta)^n$ gives

$$f(\delta) = n\delta - \frac{n(n-1)}{2}(1-\xi)^{n-2}\delta^2$$

for some $0 \le \xi \le \delta$, proving (16). On (17), $k_\sigma\delta_\sigma \to 0$, so the remainder is $o(k_\sigma\delta_\sigma r_H^{-2k_\sigma})$ and (18) follows.

The archived multiplier law is

$$\beta_k = \beta\exp\left[-\frac{\log C_M}{2k} + o(k^{-1})\right], \qquad C_M = 1.946342905200968\ldots. \tag{20}$$

Hence

$$\frac{\mathcal{A}_{k,2k}}{2k} = \frac{\beta^{2k}}{C_M}\{1+o(1)\}. \tag{21}$$

After multiplication by R^{2k}, this is a constant multiple of $(\beta R)^{2k}$. The weighted parity packet divided by $2k$ is $C_*\sqrt{\sigma}(R/r_H)^{2k}\{1+o(1)\}$. Using $2k = \log(1/\sigma)/\log\lambda + O(1)$ gives the same exponent (19). □

4 Clock phase and a scalar-only obstruction

The bounded integer phase must be retained:

$$\eta_\sigma = k_\sigma - \frac{\log(1/\sigma)}{2\log\lambda}. \tag{22}$$

Theorem 4.1 (Phase law and unique scalar balance)**.** *Along any subsequence on which $\eta_\sigma \to \eta$,*

$$\boxed{\frac{\mathcal{P}_{\sigma,2k_\sigma}}{\mathcal{A}_{k_\sigma,2k_\sigma}} \longrightarrow C_* C_M \lambda^\eta.} \tag{23}$$

The leading scalar packets balance only at

$$\eta_* = -\frac{\log(C_* C_M)}{\log\lambda} = 3.060914913721564\ldots. \tag{24}$$

For every phase with $|\eta| \le 1$,

$$C_* C_M \lambda^{\eta} \le C_* C_M \lambda = 0.343888019998872\ldots < 1. \tag{25}$$

Therefore, on such a subsequence, if $\mathcal{T}_{\sigma,2k} = o(\mathcal{A}_{k,2k})$, then

$$|e_{\sigma,k,2k}| \ge \{0.656111980001127\ldots + o(1)\}\mathcal{A}_{k,2k}, \tag{26}$$

and this is not $o(kR^{-2k})$.

Proof. Divide (18) by (21) and use $\beta^{2k} = r_H^{-2k}\lambda^{-k}$:

$$\frac{\mathcal{P}_{\sigma,2k}}{\mathcal{A}_{k,2k}} = C_* C_M \sqrt{\sigma}\,\lambda^k\{1 + o(1)\}.$$

Equation (22) makes $\sqrt{\sigma}\lambda^k = \lambda^{\eta_\sigma}$, proving (23). Monotonicity in η gives the unique balance phase and (25). Substitute the ratio into (13). Finally,

$$\frac{\mathcal{A}_{k,2k}}{kR^{-2k}} = \frac{2}{C_M}(\beta R)^{2k}\{1 + o(1)\} \longrightarrow \infty$$

because $\beta R > 1$. □

This is deliberately a scalar-only negative result. It proves that omitting an alias-scale raw boundary/shell packet cannot work on common floor/ceil/nearest clocks. It does not bound the actual raw trace packet and does not prove divergence of the combined physical residual.

The same phase controls the local clearance. If the archived boundary-cycle clearance is

$$\Delta_k^{\mathrm{clr}} = C_{\mathrm{b}}\lambda^{-2k}\{1 + o(1)\}, \qquad C_{\mathrm{b}} = 0.4608051492\ldots, \tag{27}$$

then

$$d_{\sigma,k} := \frac{\Delta_k^{\mathrm{clr}}}{\sigma} = C_{\mathrm{b}}\lambda^{-2\eta_\sigma}\{1 + o(1)\}. \tag{28}$$

At scalar balance, $d_* = C_{\mathrm{b}}(C_* C_M)^2 = 0.019340633095025\ldots$, whereas $|\eta| \le 1$ corresponds to $0.163544744607256\ldots \le d \le 1.298368749415710\ldots$. Thus suppressing the clock phase would discard both the scalar balance ratio and the RH-322 entrance profile.

5 Typed interface for the neighboring shell

RH-322–RH-324 retain the entrance clearance and the oriented affine frame

$$V \sim g_d, \qquad U = -\alpha V - Z_1, \qquad W = -\lambda U + Z_2, \qquad (V, U, W) \text{ has orientation } (+, -, +), \tag{29}$$

where $\alpha = 2u_c$ and the output is compared after the center shift $\kappa_{\mathrm{aff}} d = 2u_c\lambda d$. Their objects are forward probability laws of mass one. They do not yet define a signed trace observation, and RH-19 shows that the neighboring critical sibling cannot be hidden in a far Gaussian tail.

The exact handoff is therefore conditional but typed. A later theorem must construct, on the same clock, phase, coordinate frame, and Hardy normalization, a decomposition

$$\mathcal{T}_{\sigma,2k} = \mathcal{B}_{\sigma,k}(d, \eta; V, U, W, \mathfrak{t}) + \mathcal{S}_{\sigma,k}(d, \eta; \mathfrak{t}) + \mathcal{R}_{\sigma,k}. \tag{30}$$

Here $\mathfrak{t}$ is the still-missing trace observation, $\mathcal{B}$ is the physical boundary packet, $\mathcal{S}$ is the neighboring-shell packet, and $\mathcal{R}$ contains the second-leg, all-leg transport, and observation remainders. Combining (30) with theorem 2.2 gives the exact join

$$e_{\sigma,k,2k} = \mathcal{B}_{\sigma,k} + \mathcal{S}_{\sigma,k} + \mathcal{R}_{\sigma,k} + \mathcal{P}_{\sigma,2k} - \mathcal{A}_{k,2k}. \tag{31}$$

Consequently, if $\mathcal{R}_{\sigma,k} = o(kR^{-2k})$, the desired matching is equivalent to the signed condition

$$\boxed{\mathcal{B}_{\sigma,k} + \mathcal{S}_{\sigma,k} = \mathcal{A}_{k,2k} - \mathcal{P}_{\sigma,2k} + o(kR^{-2k}).} \tag{32}$$

Equation (32) is an interface, not a theorem that the packets on its left have been constructed.

6 Reproduction and claim boundary

The reproduction code checks the exact moment split, the parity signs, the uniform Taylor bound, the clock/clearance dictionary, and the scalar balance law. It evaluates the following constant ledger:

η	$C_* C_M \lambda^\eta$	$C_\mathrm{b} \lambda^{-2\eta}$
-1	0.122049588	1.298368749
0	0.204869205	0.460805149
1	0.343888020	0.163544745
3.060914914	1.000000000	0.019340633

These are evaluations of proved symbolic formulas, not fitted noisy traces.

Remark 6.1 (Claim boundary). The exact identity combines only known algebraic packets. It does not identify the RH-322–RH-324 probability law with $\mathcal{T}_{\sigma,2k}$, control the second physical critical leg, prove all-leg phase transport, bound the weighted trace observation, include the neighboring shell, or establish (30). Hence no joint first-alias trace law, full-trace replacement, or actual divergence theorem is obtained. Gates A–E remain false/open. This paper constructs no Hilbert–Polya operator, identifies no Riemann zero, proves no von Mangoldt trace formula or zeta-divisor equality, and does not imply RH.

RH-327 should now compute the neighboring-shell coupling in the typed frame of (30); RH-328 may formulate the joint matching equation only after that shell and the trace observation are defined.